

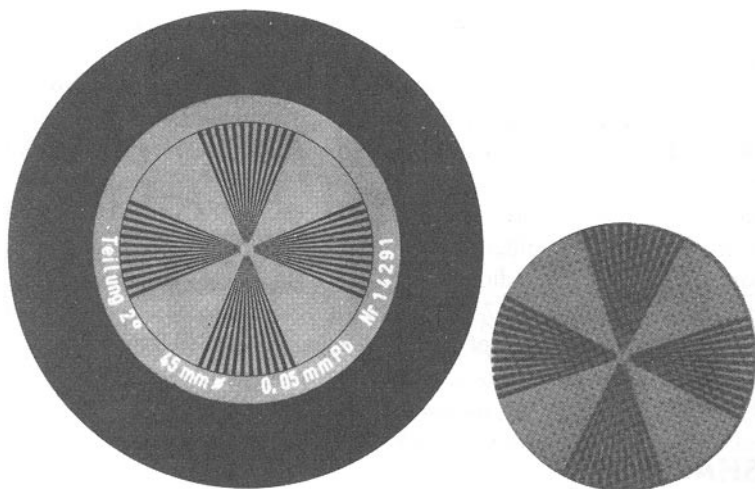
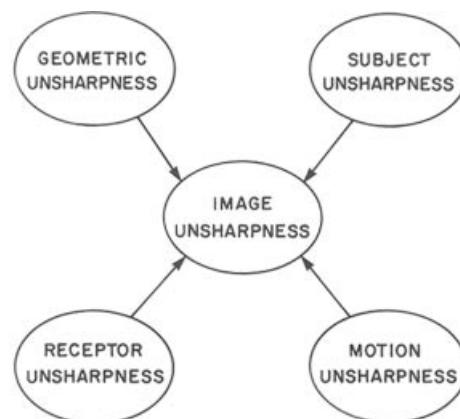
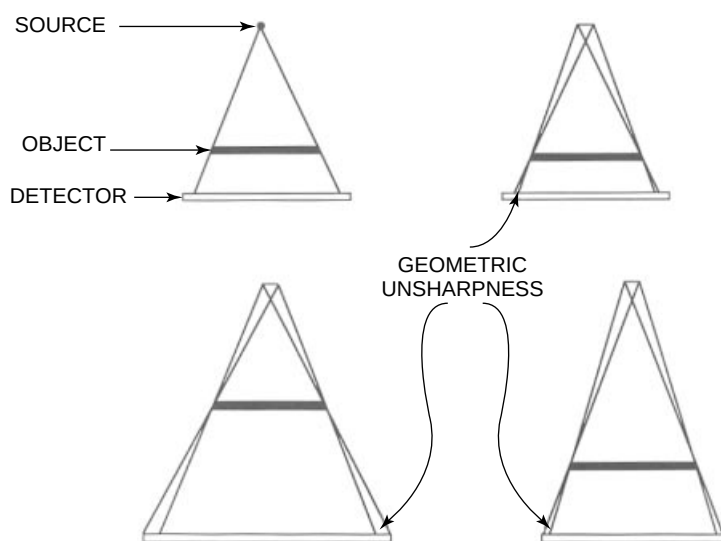
**FIGURE 16-1**

Image unsharpness is the blurring in an image of sharp edges in an object. The contact radiograph of a star test pattern on the left displays negligible unsharpness, while the x-ray image on the right reveals noticeable blurring. This is Figure 5-9, repeated here for convenience. (From Hendee, W. R., Chaney, E. L., and Rossi, R.P.⁵ Used with permission.)

**MARGIN FIGURE 16-2**

Contributions to image unsharpness.

and image receptor. These influences are depicted in Figure 16-2 for the case of x-ray imaging. With minor modifications, they are applicable to all cases of radiologic imaging. In the upper left diagram of Figure 16-2, the radiation source is very small (i.e., a conventional focal spot of 0.6 to 1 mm). In this case, the sharp borders in the object are blurred over a finite region of the image. The image presents the borders with a degree of fuzziness that increases with the size of the focal spot. The blurring also increases with the distance of the image receptor from the object, as shown in the lower left diagram of Figure 16-2. By similar reasoning, the geometric unsharpness is reduced by moving the radiation source away from the object, thereby increasing the source-to-object distance (lower right diagram in Figure 16-2).

**FIGURE 16-2**

Geometric unsharpness. **Upper left:** Minimal unsharpness with a focal spot of infinitesimal size. **Upper right:** Unsharpness contributed by a focal spot of finite size. **Lower left:** Increased unsharpness caused by moving the receptor farther from the object. **Lower right:** Reduction in unsharpness caused by moving the object farther from the radiation source and closer to the detector.

Magnified x-ray images are often acquired by placing the image receptor at some distance from the patient. This procedure lengthens the object-to-receptor distance, thereby increasing the geometric unsharpness. Usually this increase is limited by use of a small focal spot.

Geometric unsharpness sometimes is referred to as the “penumbra” or “edge gradient.”

Compared with the cathode side, the anode side of the x-ray image reveals (a) reduced unsharpness because of the line-focus principle and (b) reduced x-ray intensity because of a greater heel effect.

The relationship between the size of the radiation source and the distances between source and object and between object and image receptor (detector) can be expressed mathematically. The expression is

$$U_g = \frac{(\text{Focal spot size})(\text{Object-to-receptor distance})}{(\text{Source-to-object distance})}$$

Geometric unsharpness does not vary with the position of a boundary across an image. That is, objects near the edge of an image display the same geometric unsharpness as those near the center, provided that the projected size of the radiation source remains constant. This size is constant across the image in a direction perpendicular to the anode–cathode axis of the x-ray tube. In a direction parallel to the axis, however, the projected focal spot diminishes in size in a direction towards the anode. This effect, described earlier as the line focus principle, reduces the unsharpness on the anode side of the image compared with the cathode side.

Example 16-1

Determine the geometric unsharpness for a radiographic procedure with a 2-mm focal spot, object-to-receptor distance of 25 cm and source-to-receptor distance of 100 cm.

The source-to-object distance is $100 \text{ cm} - 25 \text{ cm} = 75 \text{ cm}$.

The image unsharpness is

$$\begin{aligned} U_g &= \frac{(\text{Focal spot size})(\text{Object-to-receptor distance})}{\text{Source-to-object distance}} \\ &= \frac{(2 \text{ mm})(25 \text{ cm})}{(75 \text{ cm})} \\ U_g &= 0.67 \text{ mm} \end{aligned}$$

Every imaging technology produces blurred images as a consequence of imaging features analogous to those described for radiography. For example, nuclear medicine images exhibit considerable unsharpness as a result of the finite diameter of the collimator holes and the thickness of the collimator. In computed tomography, geometric unsharpness is determined primarily by the size of the x-ray tube focal spot and the diameter and length of the collimators in front of the radiation detectors. In magnetic resonance imaging (MRI), image unsharpness is strongly influenced by the degree to which extracted signals can be localized to a specific region of tissue by the use of temporally and spatially varying magnetic fields. In ultrasound, image unsharpness is affected by many factors, including the width of the ultrasound beam and the distance between the transducer and reflecting boundaries separating different tissues in the body.

Subject Unsharpness

Not all structures within the patient present well-defined boundaries that can be displayed as sharp borders in the image. Often a structure is distinguishable anatomically from its surroundings only by characteristics that vary gradually over distance. Also, the shape of an object may prevent the projection of sharp boundaries onto the image receptor. This latter situation is depicted in Figure 16-3, where the object on the left presents sharp borders in the image and the object on the right is depicted with blurred edges. Image unsharpness that is contributed by the object is known as subject (object) unsharpness. It can be a result of the composition of the object, its shape, or a combination of both.

Motion Unsharpness

Motion is often a major contributor to unsharpness in a radiologic image. Usually the motion occurs in the anatomic region of interest as a result of involuntary physiologic processes or voluntary actions of the patient. Motion causes boundaries in the patient to be projected onto different regions of the image receptor while the image is being

Subject unsharpness is sometimes referred to as “absorption unsharpness.” Subject unsharpness may be described quantitatively with the expression

$$U_s = \frac{D_2 - D_1}{x}$$

where $D_2 - D_1$ is the difference in optical density or brightness over a boundary of linear dimension x ⁶.

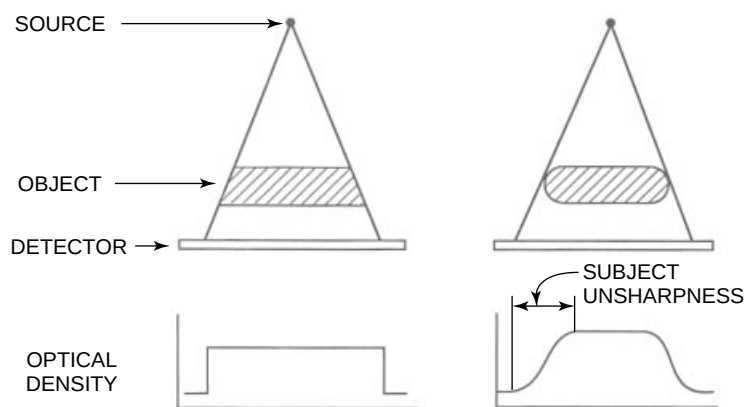


FIGURE 16-3

Subject unsharpness. Edges of the trapezoid on the left are parallel to the path of x-rays, and the resulting image reveals no subject unsharpness. The ellipsoid on the right yields an image of varying density from the edges to the center and presents the observer with substantial subject unsharpness.

formed. As a result, the boundaries are spread over a finite distance, and the resulting borders are blurred in the image.

Voluntary motion often can be controlled by keeping examination times short and asking the patient to remain still during the examination. However, these exhortations sometimes are ineffective, especially when the patient is an infant, demented, or in pain. In some cases, a family member can hold or soothe the patient and reduce voluntary motion during the examination. If the family member is exposed to radiation, protective garments should be provided to reduce the exposure. Occasionally, physical restraints and anesthetics may be required.

Involuntary motion is a more frequent problem and is especially troublesome when images of fast-moving structures such as the heart and great vessels are required. For many regions, motion can be “stopped” in the image by the use of very short examination times. In chest images, for example, examination times of a few milliseconds are used to gain a reasonable picture of the cardiac silhouette without the perturbing influence of heart motion. Short examination times are also desired in studies of the gastrointestinal tract to reduce motion unsharpness caused by peristalsis. In fact, short examination times are a general rule in radiography because voluntary and involuntary motion is invariably present to some degree when any anatomic region is being imaged.

Every image receptor requires a certain level of radiation exposure to produce an acceptable image. Some receptors are more sensitive to radiation than others and require less exposure. In radiography, for example, fast intensifying screens require less exposure than do detail screens, and shorter exposure times can be used. But fast screens produce more receptor unsharpness (discussed below), so the reduction in unsharpness achieved by using fast screens may be compromised by the greater receptor unsharpness contributed by the screens. Also, delivery of the required amount of radiation to a given set of screens must occur at higher intensity if short exposure times are used. Hence, higher tube currents are required with short exposure times so that the mA·s yields sufficient radiation to expose the screens properly. Higher tube currents (mA) require larger focal spots that produce greater geometric unsharpness in the image. In summary, every radiographic examination represents a trade-off among geometric, motion, and receptor unsharpness, with the goal of minimizing the total unsharpness in the image. This goal is achieved only by thoughtful selection of the tools and techniques of the examination based upon their characteristics and contributions to the image.

Receptor Unsharpness

The image receptor collects data generated during the imaging process and displays it as a gray-scale or color image. In some techniques such as conventional radiography,

Typical Speeds of Anatomic Structures that Cause Blurring of the Image

Extremities	0.5–1 mm/sec
Head	1–2
Upper abdomen	20–40
Lower abdomen	15–30
Adult lungs	75–100
Infant lungs	150–200
Adult heart muscle	75–100
Infant heart muscle	150–200
Heart valves	400–500
	~50 mi/hr

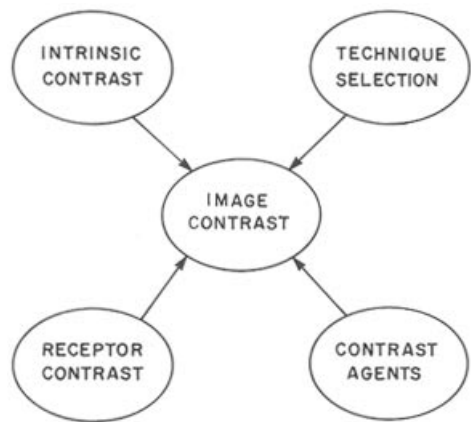
The most frequent causes of motion unsharpness in x-ray images are heartbeat, the pulse, peristalsis, and the uncooperative patient.

The radiographer should hold a patient during exposure only under extraordinary circumstances when no adult family member is available.

The evolution of x-ray tubes, including Coolidge’s hot cathode, the rotating anode, and specially fabricated composites, reflects the nonending quest for shorter, higher-intensity exposures.

In situations where image clarity is noise-limited, as discussed later in this chapter, receptor unsharpness sometimes can be used to reveal structures that are otherwise hidden (i.e., “buried in the noise”). The same effect can be achieved by defocusing the eyes slightly while viewing a noise-limited image.

The sensitivity of an x-ray receptor is the probability that an x ray will interact in the receptor. This probability is termed the *quantum detection efficiency* n .⁷ This parameter is defined as $n = 1 - e^{-\mu x}$, where μ is the linear attenuation coefficient of the receptor material and x is the thickness of the receptor.



MARGIN FIGURE 16-3
Contributions to image contrast.

In medical imaging of soft tissues, contrast is frequently the image characteristic of greatest concern.

The contrast between adjacent regions in an image can be described quantitatively as $D_2 - D_1$, the difference in optical density or brightness between the two regions.

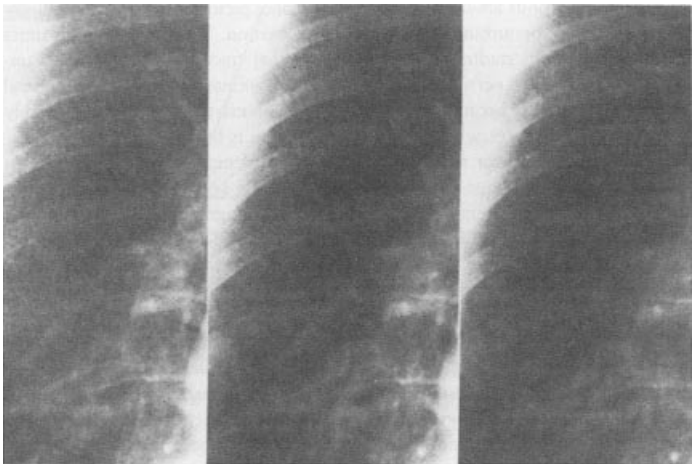


FIGURE 16-4
In this digital chest radiograph, multiple pulmonary nodules are displayed at three levels of spatial resolution. The finer display yields reduced receptor/display unsharpness.
Left: 0.2 × 0.2-mm pixels. **Center:** 0.4 × 0.4-mm pixels. **Right:** 0.6 × 0.6-mm pixels.
(From Foley, W. D.⁸ Used with permission.)

the receptor converts x-ray data to an image directly by the use of relatively simple devices such as intensifying screens and film. In other imaging methods the conversion process is more complex and employs a computer to form an image on a video display. In every display technique, no matter how simple, the image receptor inevitably adds unsharpness to the image. This contribution to image unsharpness is termed *receptor unsharpness*.

In radiography, receptor unsharpness is determined principally by the thickness and composition of the light-sensitive emulsion of the intensifying screens. These characteristics influence not only receptor unsharpness but also the sensitivity of the screens to x rays. With increasing thickness, for example, the sensitivity improves and the unsharpness increases. The choice of screens is, consequently, a trade-off between unsharpness introduced by the receptor and that resulting from motion caused by the finite time to record the imaging data.

In many imaging techniques, the technique of image display can influence the overall unsharpness of the image. In digital radiography, for example, the display device can provide different levels of unsharpness. In Figure 16-4 the same data are presented in three display formats. In the image on the left the data are presented in a fine matrix of 0.2 × 0.2-mm picture elements (pixels), and the image appears quite continuous and smooth. On the right the same data are displayed in a coarser matrix (0.6 × 0.6-mm pixels), and the image has a chunkier, rather “blocky” appearance. In this example, the coarser display adds significant unsharpness to the image.

CONTRAST

Contrast is the second major feature of an image. This characteristic describes how well the image distinguishes subtle features in the object (patient). Image contrast is depicted in Figure 16-5. In diagnostic imaging, the contrast of an image is a product of complex interactions among the anatomic and physiologic attributes of the region of interest, the properties of the imaging method and receptor employed, and conscious efforts to influence both the intrinsic properties of the region and its presentation as an image. These interactions are characterized as four influences on image contrast, as depicted in the margin.

Intrinsic Contrast

The underlying philosophy of diagnostic imaging is that structures in the patient can be distinguished in an image because they differ in physical composition and



FIGURE 16-5
Different levels of image contrast. (From Hendee, W. R.⁴ Used with permission.)

physiologic behavior. These differences are referred to as intrinsic (sometimes called “subject,” “object,” or “patient”) contrast. Physical properties of the patient that contribute to intrinsic contrast are depicted in Table 16-1.

In radiography, intrinsic contrast is a reflection primarily of atomic number and physical density differences among different tissues. Some structures (e.g., breast) exhibit very subtle differences in composition and are said to have low intrinsic contrast. Other structures (e.g., chest) provide large differences in physical density and atomic composition and yield high intrinsic contrast.

Imaging Technique

Every imaging application reflects a choice of a specific imaging technique and receptor among many alternatives to yield images of greatest potential to provide the desired information. For a specific imaging application, image contrast can be influenced by careful selection of the technique factors used to produce the image. In radiographic imaging of the breast, for example, subtle differences among tissues can be accentuated in the image by use of low peak kilovoltage and small amounts of beam filtration. These choices enhance the differential transmission of x rays through tissues that vary slightly in atomic composition and physical density. In chest radiography, relatively high peak kilovoltage and large amounts of beam filtration are chosen to suppress differential absorption of x rays in bone so that lesions of increased physical density can be detected in the lung parenchyma under the ribs. The choice of pulse sequence and other variables in MRI strongly influences the resulting contrast among structures in the image. This influence is shown in Figure 16-6 for the same anatomic region imaged by different pulse sequences.

Contrast Agents

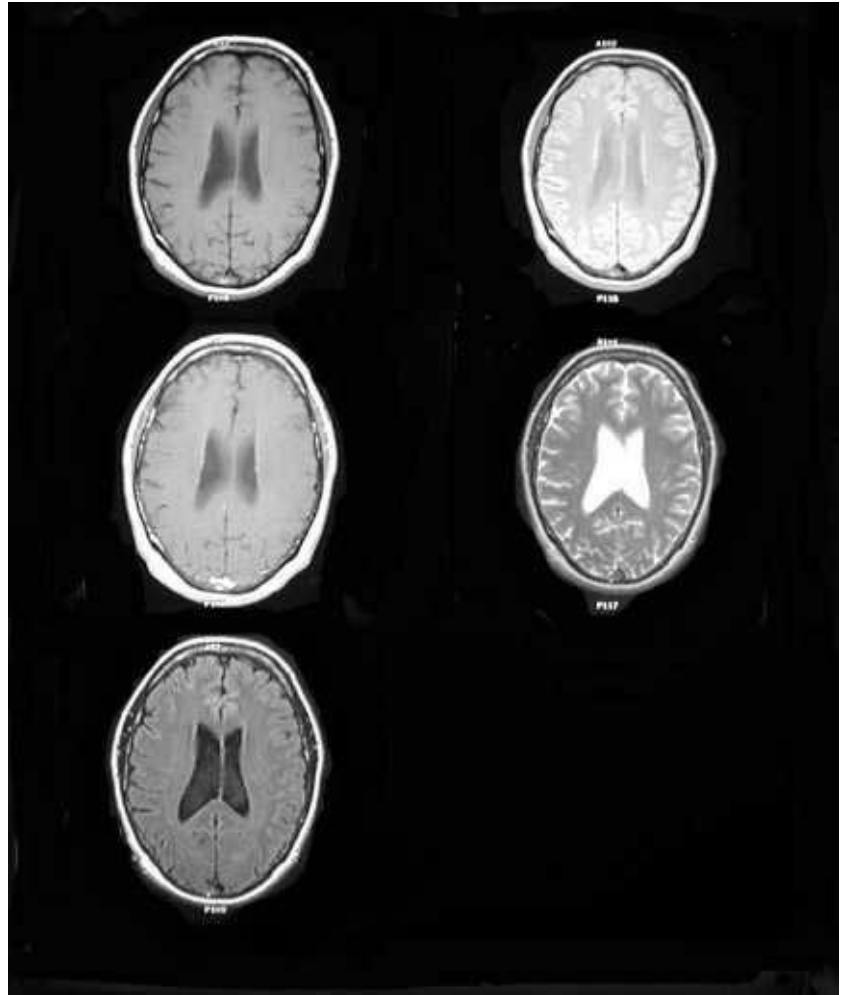
In some radiologic examinations a substance can be introduced into the body to enhance tissue contrast. This substance, termed a contrast agent (or contrast medium or dye), is selected to provide a signal different from that of the surrounding tissues. In angiography, a water-soluble agent containing iodine is injected into the circulatory system to displace blood and thereby enhance the contrast of blood vessels by increasing the attenuation of x rays impinging on the vessels. This technique can be further distinguished as arteriography or venography, depending on which side of

In MRI, a number of tissue properties contribute to the imaging data, and the type of pulse sequence and its parameters are chosen to provide a desired level of contrast among tissues.

Low kVp and small amounts of beam filtration yield a relatively “soft” x-ray beam with a significant likelihood of photoelectric interactions. High kVp and large amounts of beam filtration yield a relatively “hard” x-ray beam that interacts almost entirely by Compton interactions.

TABLE 16-1 Physical Properties that Influence Intrinsic Contrast for Various Imaging Techniques

<i>Radiography</i>	<i>Nuclear Medicine</i>	<i>Ultrasound</i>	<i>Computed Tomography</i>	<i>Magnetic Resonance</i>
Atomic number	Activity	Velocity	Physical density	Proton density
Physical density	Distribution	Physical density	Electron density	Relaxation times
Thickness	Thickness	Thickness	Flow	Flow

**FIGURE 16-6**

Representative magnetic resonance images of the normal brain

UL T₁ weighted image

ML T₁ weighted image with contrast

LL FLAIR (Fluid attenuated inversion recovery) image

UR Proton density image

LR T₂-weighted image

Source: Courtesy of David Daniels, M.D., Medical College of Wisconsin, Department of Radiology

the circulatory system the injection is made. The iodine attenuates x rays strongly because its K-absorption edge of 33 keV is ideally positioned with respect to the energy distribution of x rays typically employed for diagnosis.

Several other types of contrast agents are used in diagnostic imaging. For example, a contrast agent can be used to displace the cerebrospinal fluid in studies of the central nervous system. For studies of the spinal cord, the contrast agent may be “ionic” or “nonionic,” and the procedure is termed “myelography.” Contrast agents are useful in computed tomography as well as in radiography. Iodine-containing compounds are also helpful in studies of the kidneys and urinary tract, where the applications are known as urography. Compounds containing barium are frequently used for radiographic examination of the gastrointestinal tract, with the barium compound administered orally (barium swallows) or rectally (barium enemas). Solutions containing dissolved “microbubbles” of carbon dioxide are sometimes employed as contrast agents in ultrasound, and gadolinium and other substances are increasingly being used as contrast agents in MRI because of their influence on the relaxation constants of tissues.

New contrast agents, especially for MRI¹⁰ and ultrasound,¹¹ are improving diagnostic imaging. With the arrival of nonionic agents for radiography and computed tomography, the risk of occasional reactions to contrast agents was reduced. These newer agents were initially more expensive and raised the issue of increased cost for a slight reduction in risk.¹² As the use of nonionic contrast agents has increased, the cost differential between ionic and nonionic agents has diminished and is no longer a major impediment to the use of nonionic agents.

Receptor Contrast

The contrast of an image is affected by the properties of the receptor used to form the image. In radiography, the choice of film has a substantial influence on image contrast. Observers who prefer a high-contrast image tend to select film that yields a steep characteristic (H-D) curve with high contrast and narrow latitude. Those who prefer less contrast and wider latitude of exposure choose a film with a more gradual slope for the characteristic curve.

Display of images on a video monitor adds flexibility in the choice of image contrast because the viewer can alter the receptor contrast. Just as with film, however, the dynamic range of the monitor is limited, and choosing a high-contrast setting limits the ability to display a wide range of exposures.

Increased flexibility in altering image contrast is one of the major advantages of digital imaging. When imaging data are available in digital form, the viewer can select any range of exposures for display in a gray-scale image. The range can be altered at will and the result captured on film if desired while the total range of data is preserved in computer storage. Shown in Figure 16-7 are displays at different contrast settings for the same set of computed tomographic data. Enhanced contrast and reduced exposure range are apparent in the images obtained with narrower windows.

When film is the image receptor, image contrast is highly dependent on the conditions under which the film is processed. A high level of quality control over film processing is essential to the production of acceptable film-based medical images.

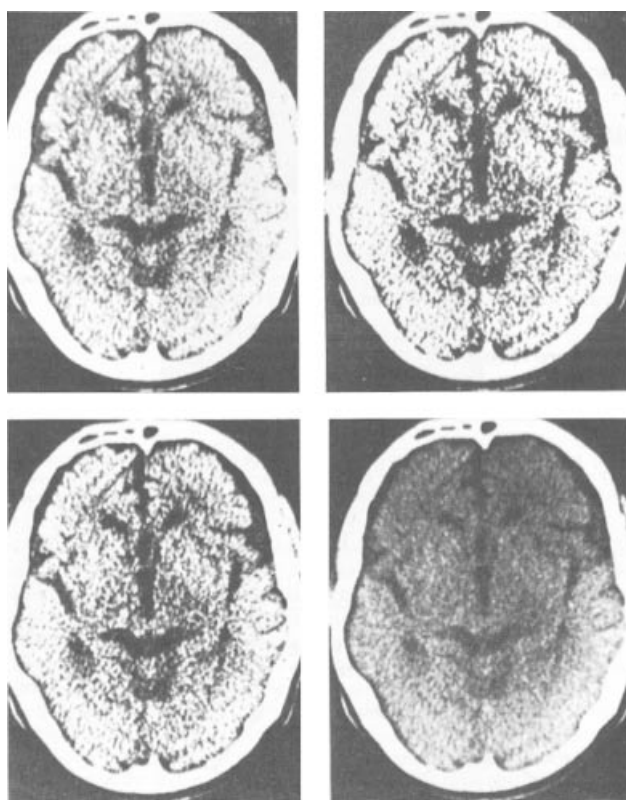
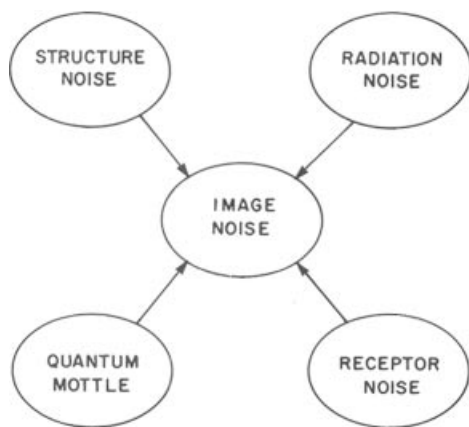


FIGURE 16-7

Computed tomographic data displayed with a constant window level but with different settings of window width. (From Hendee, W. R.¹³ Used with permission.)

Noise in an imaging system can be demonstrated by exposing the system to a uniform fluence of radiation (e.g., x rays). The macroscopic variation in optical density or brightness about an average value is a result of the “noise” contributed by the imaging system.



MARGIN FIGURE 16-4
Contributions to image noise.

The influence of radiation noise is often described by the signal-to-noise ratio (SNR), defined as:

$$\begin{aligned}\text{SNR} &= nN_0/[nN_0]^{1/2} \\ &= (nN_0)^{1/2}\end{aligned}$$

where n is the quantum detection efficiency and N_0 is the number of incident photons.

■ IMAGE NOISE

Every radiologic image contains information that is not useful for diagnosis and characterization of the patient's condition. This information not only is of little interest to the observer, but often also interferes with visualization of image features crucial to the diagnosis. Irrelevant information in the image is defined as image noise. Image noise has four components: structure noise, radiation noise, receptor noise, and quantum mottle.¹⁴

Structure Noise

Information about the structure of the patient that is unimportant to diagnosis and characterization of the patient's condition is known as structure noise. For example, shadows of the ribs not only are irrelevant in chest radiographs taken to examine the lung parenchyma, but also can hide small lesions in the parenchyma that could be important to characterization of the patient's condition. However, rib shadows are important to the diagnosis of a cracked rib, and in this case the parenchyma and mediastinum could be characterized as structure noise. Structure noise is defined not only by the information present in the image but also by the reasons why the image was obtained in the first place.

Structure noise is one of the more disturbing features of radiologic images and is often responsible for missed lesions and undetected abnormalities. In radiography, patients frequently are positioned for examination so that structure noise is reduced in the region of interest in the image. The most obvious method of reducing structure noise is tomography, which blurs structures above and below the plane of interest within the patient. Tomographic images are obtained automatically in computed tomography and MRI, and the superior rendition of low-contrast structures by these technologies is attributable in part to their reduced level of structure noise.

Radiation Noise

Radiation noise is information present in a radiation beam that does not contribute to the usefulness of the image. For example, many x-ray beams exhibit a nonuniform intensity from one side to the other. This nonuniformity, termed the *heel effect*, is a result of the production and filtration of the beam in the anode of the x-ray tube. Although the nonuniformity provides information about these mechanisms, that information is irrelevant to diagnosis of the patient's condition. Similarly, the radiation beam exiting from the patient contains considerable scattered radiation. This radiation contains information about the patient, although in such a complicated form that it cannot be easily decoded. As it impinges on the image receptor, scattered radiation interferes with the visualization of patient anatomy. Hence, scattered radiation is radiation noise. In projection radiography, the contribution of scattered radiation to image noise is so severe that it must be removed by a grid to permit the structures of interest to be seen clearly.

Radiation noise is present with every imaging technique, but is sometimes not very troublesome. In computed tomography, for example, the thin sections exposed during the acquisition of imaging data do not yield very much scattered radiation. In nuclear medicine, scattered radiation can be rejected by use of a pulse height analyzer because all scattered photons have energies less than the primary γ rays. In ultrasound, scattered radiation sometimes contributes to a “speckle pattern” in the image that aids in the diagnosis of certain conditions.

Receptor Noise

Most image receptors are not uniformly sensitive to radiation over their active surfaces. These receptors impose a pattern of receptor noise onto the image. In radiography,



FIGURE 16-8

Contamination on an intensifying screen is a common contributor to receptor noise. (From Hendee, W. R.⁴ Used with permission.)

intensifying screens and film are fairly uniform in their sensitivity to radiation, and their contribution to image noise is usually undetectable. However, contamination on a screen can add substantial noise to the image, as shown in Figure 16-8. In some imaging methods, receptor noise can be a major problem. In nuclear medicine, for example, unbalanced detectors can contribute significant receptor noise, as shown in Figure 16-9. In most scintillation cameras, minor variations in sensitivity among detectors are compensated automatically by computer monitoring and correcting. In MRI, surface coils contribute receptor noise through nonuniform patterns of sensitivity.¹⁶

Increasingly, imaging data are processed through computer algorithms to yield displayed images. Imperfections in the algorithms can produce imperfections in the images. These imperfections are an additional form of receptor noise. They are encountered most often with reconstruction algorithms used in CT, MRI, SPECT, and PET imaging.

Quantum Mottle

In many imaging methods, image noise is dominated by quantum mottle resulting from the finite number of information carriers (x rays, γ rays, etc.) used to create

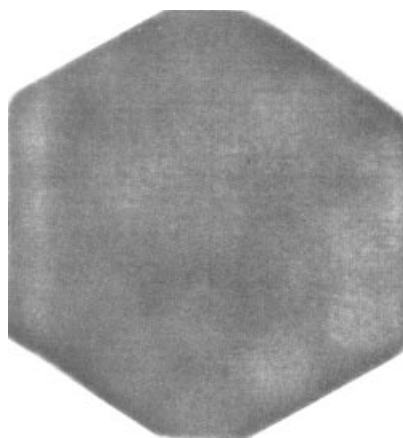


FIGURE 16-9

Unbalanced detectors can produce considerable noise in nuclear medicine images. (From Adams, R.¹⁵ Used with permission.)

**FIGURE 16-10**

Quantum mottle. As the illumination of the object increases, quantum mottle decreases, and the clarity of information improves. (From Rose, A.¹⁷ Used with permission.)

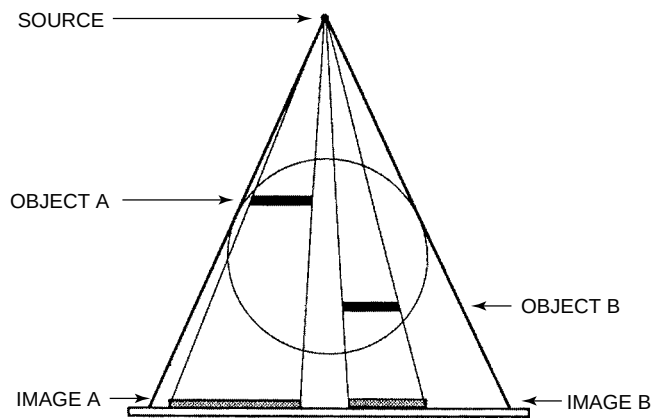
the image. Quantum mottle is illustrated in Figure 16-10. This contribution to image noise can be particularly disturbing in images of relatively low contrast. Quantum mottle can be reduced only by using more information carriers to form the image. In most cases, more information carriers are provided only at the expense of longer imaging times and increased radiation dose to the patient.

■ IMAGE DISTORTION AND ARTIFACTS

Distortion and artifacts are present to some degree in every diagnostic image. Image distortion is caused by unequal magnification of various structures in the image. In radiography, for example, structures near the intensifying screens are magnified less than those farther away, as shown in Figure 16-11. The experienced interpreter learns to use these differences in magnification to gain a perspective of depth among different structures in the image.

Magnification M in an image is the ratio of image to object size. In radiography, magnification can be computed as:

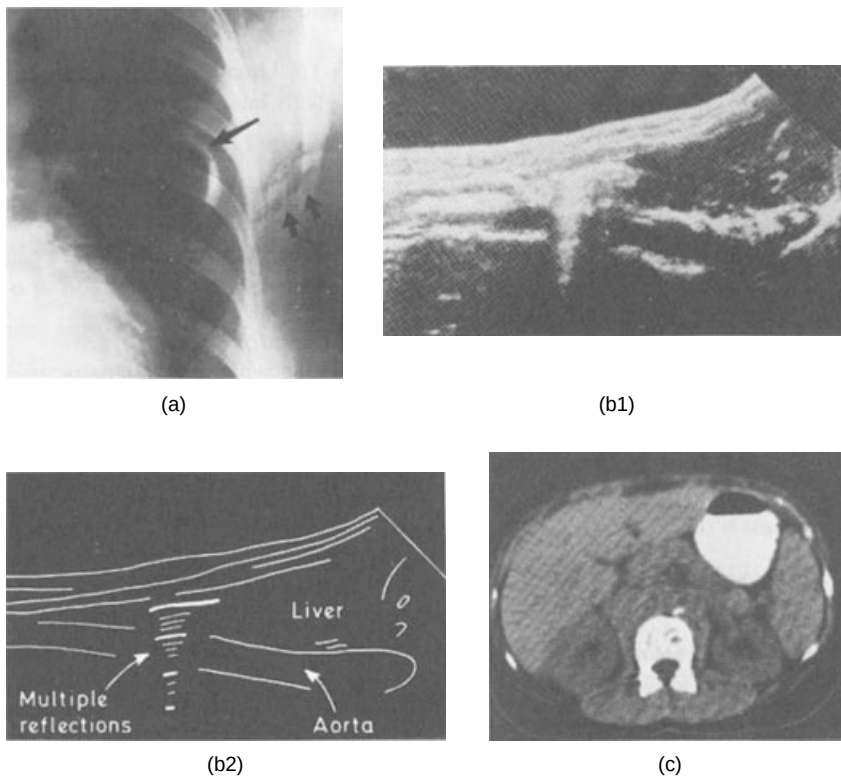
$$\begin{aligned}
 M &= \frac{\text{Image size}}{\text{Object size}} \\
 &= \frac{\text{Source-to-receptor distance}}{\text{Source-to-object distance}}
 \end{aligned}$$

**FIGURE 16-11**

Distortion is the unequal magnification of different structures recorded in an image.

Artifacts can arise from so many causes that their complete description is not possible here. Examples include reverberation artifacts in ultrasound, streaks caused by moving structures of high density in computed tomography, and artifacts produced by nonuniformities in the magnetic field introduced by metallic structures (e.g., bridge-work) in MRI.

Examples of artifacts are shown in Figure 16-12. Usually these artifacts are easily recognizable and do not interfere greatly with interpretation of the image.

**FIGURE 16-12**

Examples of image artifacts. **A:** Radiographic artifacts caused by crinkling the film (*single arrow*) and exerting pressure on the film (*double arrows*). (From Sweeney, R. J.¹⁸ Used with permission.) **B:** Reverberation artifacts caused by multiple reflections in ultrasonography. (From McDicken, W. R.¹⁹ Used with permission.) **C:** Ring artifacts caused by imbalanced detectors in computed tomography. (From Morgan, C. L.²⁰ Used with permission.)

Example 16-2

For an image of 10 cm obtained at a source-to-receptor distance of 100 cm and an object-to-receptor distance of 20 cm, what is the size of the object?

The source-to-object distance is 100 cm – 20 cm = 80 cm.

$$\begin{aligned}
 M &= \frac{\text{Image size}}{\text{Object size}} = \frac{\text{Source-to-receptor distance}}{\text{Source-to-object distance}} \\
 \text{Object Size} &= \text{Image Size} \left[\frac{(\text{Source-to-object distance})}{(\text{Source-to-receptor distance})} \right] \\
 &= 10 \text{ cm} \left[\frac{80 \text{ cm}}{100 \text{ cm}} \right] \\
 &= 8 \text{ cm}
 \end{aligned}$$

SUMMARY

Image unsharpness is influenced by four factors:

- Geometric unsharpness
- Subject unsharpness
- Motion unsharpness
- Receptor unsharpness

Image contrast is influenced by four factors:

- Intrinsic contrast
- Image technique
- Contrast agents
- Receptor contrast

Image noise is influenced by four factors:

- Structure noise
- Radiation noise
- Receptor noise
- Quantum mottle

Image distortion is caused by unequal magnification of structure in the object. Image artifacts can arise from a multitude of conditions.

PROBLEMS

*16-1. Identify the four factors that influence the clarity of information in a diagnostic image.

*16-2. Identify the four contributions to image unsharpness.

*16-3. Estimate the total unsharpness in an image when the geometric unsharpness is 0.2 mm, motion unsharpness is 0.3 mm, subject unsharpness is 0.1 mm, and receptor unsharpness is 0.2 mm.

*16-4. For an object–receptor distance of 10 cm and a source–object

distance of 90 cm, the geometric unsharpness is 0.2 mm. Determine the focal spot size.

*16-5. For a focal spot size of 1.0 mm and fixed source–object distance of 80 cm, what object–film distance would yield a geometric unsharpness of 0.3 mm?

*16-6. Identify the four factors that contribute to image contrast.

*16-7. Name the four contributions to image noise.

*For those problems marked with an asterisk, answers are provided on p. 492.

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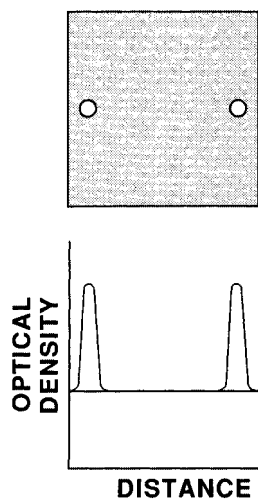
ANALYTIC DESCRIPTION OF IMAGE QUALITY

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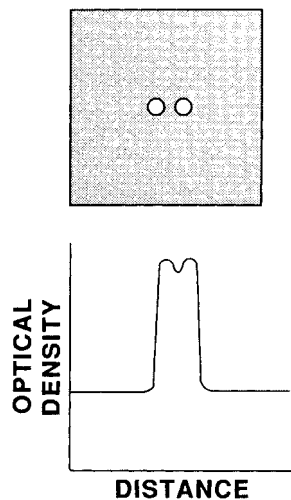
Image clarity is an objective evaluation of the ability of the image to reveal fine anatomic detail.

Image quality is the subjective response of the observer to information in the image, and it incorporates not only the physical features of the image but also the perceptual properties of the observer responding to the information.



MARGIN FIGURE 17-1

When two point-like, high-contrast objects are widely separated they can be seen as separate entities in the image.



MARGIN FIGURE 17-2

When two point-like, high-contrast objects are close together, their unsharp boundaries overlap, and they are interpreted as a single object rather than as separate objects.

The distance between the objects required for their recognition as two separate entities is the *resolution distance*.

OBJECTIVES

After studying this chapter, the reader should be able to:

- Define and distinguish among the
 - Point response
 - Line response
 - Edge response
 - Contrast response
- Describe the properties and applications of the modulation transfer function.
- Discuss the principles of quantum levels, conversion efficiencies, and the quantum sink.

INTRODUCTION

The clarity of information in a medical image is a measure of how well the image conveys information about the patient. If the information is clear and unambiguous, the clarity is acceptable. If the information is blurred or distorted, or if the image lacks contrast or is too noisy, the clarity is compromised and the information is less helpful. In these situations, the image is said to be of “unsatisfactory quality.” Image quality is a subjective expression that connotes the observer’s reaction to the clarity of information in the image. Image clarity is discussed in this chapter. The perception characteristics of the observer and how they influence image interpretation are covered in the next chapter.

POINT RESPONSE

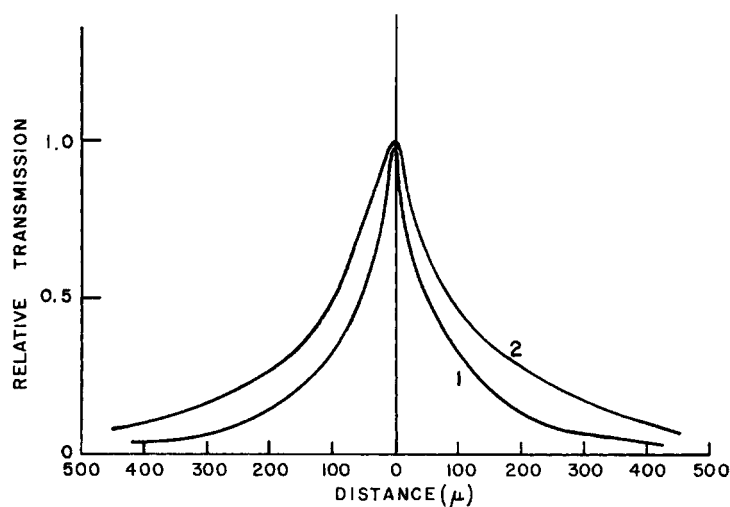
The clarity of information in an image is a measure of how well the image reveals fine structure in the object. One approach to evaluating this ability is to place two structures close together and determine how clearly they can be distinguished in the image. The structures should be very small so that they essentially constitute “point objects.” Furthermore, they should differ substantially in composition from the surrounding medium so that they have high subject contrast. Pinpoint holes in an otherwise x-ray-opaque object constitute an excellent set of point objects.

When the point objects are far apart, they are easily distinguishable in the image. As they are moved closer, their unsharp boundaries begin to overlap in the image, and the contrast diminishes between the objects and their intervening space. Eventually the diminution in contrast is so severe that the objects lose their separate identity, and they begin to be seen as a single larger objective. A difference of 10% or more in optical density usually is required to yield enough contrast for the objects to be visually separable.

A simple but rather impractical way to describe the spatial resolution of an imaging system is by use of the point spread (PSF). The PSF is an equation that mathematically describes the image of an infinitesimal point-like object (see margin). A narrow, sharply peaked PSF indicates high spatial resolution, whereas a broad, flat PSF delineates low spatial resolution.

LINE RESPONSE

A hypothetical use of point-like objects is a convenient way to introduce the concept of spatial resolution and resolving power. However, this approach is impractical from an experimental point of view because the objects need to be infinitesimally small for the approach to be valid. But very small objects such as pinpoint holes transmit very

**FIGURE 17-1**

Two normalized graphs of a line source illustrate the blurring introduced by the imaging system. System 1, obtained with x rays detected with medium-speed intensifying screens, yields better spatial resolution than system 2, which employs fast intensifying screens.

few photons, so it is difficult to distinguish them from their surroundings. A better approach to describing spatial resolution is use of the “line response.”

A line response is obtained by producing an image of a structure shaped as a line. A common object for producing a line response in radiography is an x-ray-opaque sheet that contains a narrow slit through which x-rays are transmitted.¹ A narrow catheter containing radioactive material satisfies the same purpose in nuclear imaging. The image of the line source is blurred, as shown in Figure 17-1. The amount of blurring reflects the resolving power of the imaging system. The two graphs in the illustration depict two different imaging systems, with system 1 yielding images with greater spatial resolution.

Line objects have a specific advantage over point objects for determining the resolution characteristics of imaging systems. The advantage is that line objects transmit many more x rays, and therefore the precision of the evaluation is not restricted by the number of photons used in its determination. By use of a line object, the resolving power of an imaging system can be measured under conditions of high contrast with enough photons to eliminate quantum mottle as a significant factor affecting the accuracy of the determination.

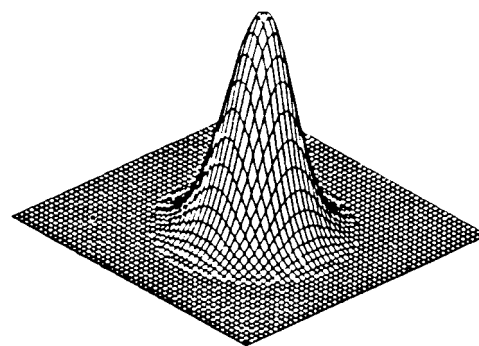
A line response such as that depicted in Figure 17-1 can be described mathematically as well as displayed pictorially. The mathematical description is known as a line spread function (LSF).² The LSF is the PSF integrated over one dimension. It is widely used in diagnostic imaging as input data for computation of the modulation transfer function (MTF), discussed later in this chapter.

The edge response is a concept that is closely related to the line response. In this approach, the imaging system is presented with a source that transmits radiation on one side of the edge and attenuates it completely on the other side. A microdensitometric scan of the resulting image yields the edge response.³

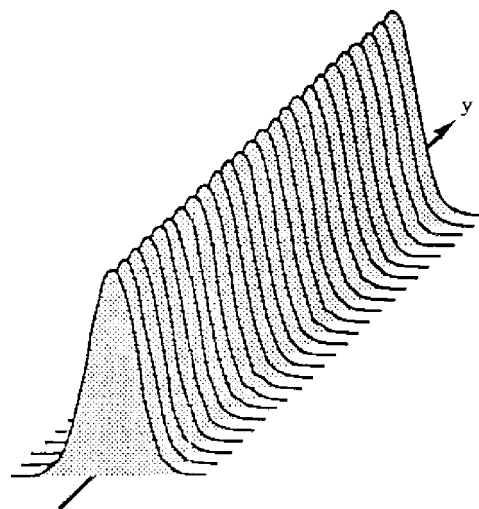
CONTRAST RESPONSE

One approach to determination of the resolution characteristics of imaging systems is to use a test object containing a series of slits separated by x-ray-opaque spaces (Margin Figure 17-5A). The slits are widely separated on one end of the object and become increasingly close together toward the other end. The transmission of x rays through this object is depicted as a square-wave pattern of reduced spatial separation (i.e., higher spatial frequency) toward the end of the object where the slits

The resolution distance is often said to reflect the *spatial resolution* of the imaging system. A system that yields excellent spatial resolution is said to have a high *resolving power*.

**MARGIN FIGURE 17-3**

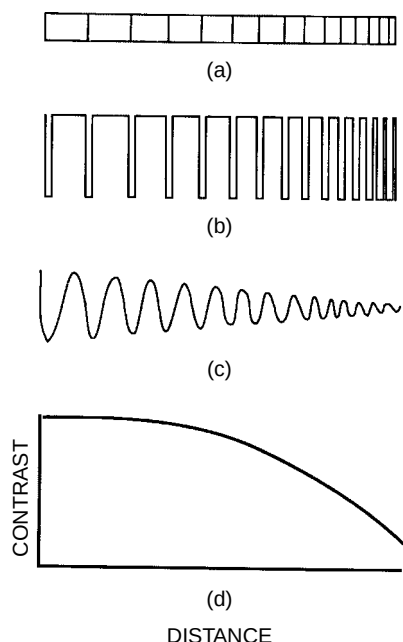
The point spread function PSF (x, y).

**MARGIN FIGURE 17-4**

The line spread function LSF (x). The derivative of the LSF is the PSF.

If the spatial resolution of the image is the same at all locations and orientations, then the PSF and LSF are said to be *spatially invariant*.

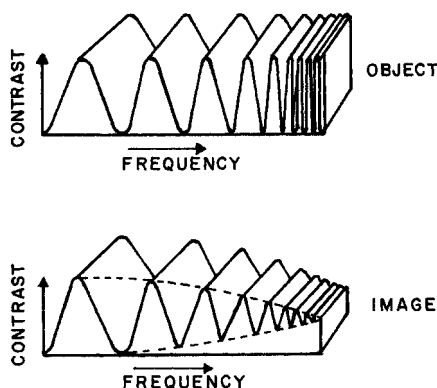
The edge spread function (ESF) is an equation that mathematically describes the shape of the edge response. The derivative of the ESF is the LSF.

**MARGIN FIGURE 17-5**

Depiction of contrast response. **A:** Test object. **B:** X-ray transmission; **C:** X-ray image. **D:** Contrast response curve.

Actually, cycles and line pairs are not valid units, and the scientifically correct expression for spatial frequency is $1/(\text{unit length})$ (e.g., $1/\text{millimeter}$), just as the correct unit for temporal frequency is $(1/\text{unit time})$ and not $(\text{cycles}/\text{unit time})$.

The difference between subject contrast in the object and the contrast presented in the image is illustrated in Margin Figure 17-6 as a function of spatial frequency.

**MARGIN FIGURE 17-6**

Suppression of contrast in the image with increasing frequency of variation in the transmission of x rays through an object.

are closer together (Margin Figure 17-5B). The spatial frequency is commonly expressed in units of cycles or “line pairs” per unit length (e.g., cycles or line pairs per millimeter).

An image of transmitted x rays is unable to maintain the square-wave pattern because the imaging system introduces unsharpness that blurs the edges (Margin Figure 17-5C). This blurring does not interfere with appreciation of separate slits when the slits are widely separated. As the slits become closer together, however, they can no longer be distinguished as separate in the image. In this region of higher spatial frequencies, the images of the slits begin to run together, and the differences between the optical densities diminish. At even higher spatial frequencies, the image is a uniform grayness, and the individual-slits and intervening spaces are undetectable.

The contrast response curve (Margin Figure 17-5D) is a single curve that depicts the amplitude of response in the image as a function of spatial frequency. The amplitude of response is the variation in optical density between the image of a slit and the image of the adjacent x-ray-opaque space.

The contrast response curve can be described mathematically as well as pictorially. The mathematical description, known as the contrast response function, is an equation that describes the amplitude of response as a function of spatial frequency.

MODULATION TRANSFER FUNCTION

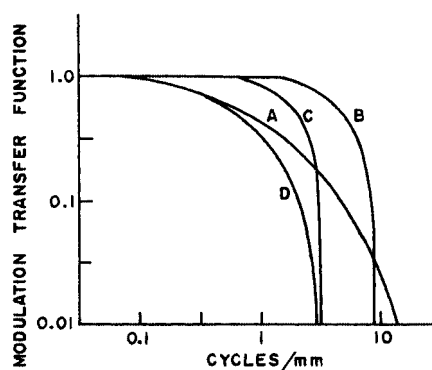
The contrast response curve and contrast response function have one significant limitation. They require a test object of alternating slits and spaces that increase in spatial frequency from one end to the other. For evaluation of a radiographic unit, frequencies as high as 10 to 15 cycles per millimeter may be required. Test objects with frequencies this high are difficult to manufacture and deploy, and they are not routinely used.

An x-ray beam transmitted through any object can be represented as a series of sine waves, with each wave possessing a characteristic frequency and amplitude. The transmitted x-ray beam is said to be modulated by the object because the distribution of amplitudes and frequencies is influenced by the object, and hence contains information about the features of the object. The function of an imaging system is to translate the modulation of the transmitted x-ray beam into a visible image and to reproduce the modulation faithfully so that features of the object are apparent. The ability of an imaging system to fulfill this responsibility is described by the modulation transfer function (MTF). That is, the MTF of an imaging system is a measure of how well sine waves that describe the transmission of x rays through an object are represented faithfully in the image.^{4,5} The MTF for an imaging system can be calculated from the PSF or LSF measured with the test objects described earlier. A curve that represents the MTF as a function of spatial frequency is shown in Margin Figure 17-7.

In the illustration, the value of the MTF is unity (i.e., 100%) at low spatial frequencies, signifying that the imaging system reproduces low frequencies without distortion or loss of resolution. As the frequency increases, the MTF decreases until it reaches zero, signifying that the spatial frequencies are so high that the imaging system provides no reproduction at all.

The MTFs for two imaging systems are shown in MF 17-8. System 1 provides a higher MTF at every spatial frequency, reflecting greater spatial resolution for this imaging system.

The MTF is a useful descriptor of the spatial resolution of an imaging system. Other characteristics of the system are also important, including its ability to reveal subtle differences in subject contrast and to provide images with low levels of noise. The MTF provides information about the clarity of information in images furnished by an imaging system, but it is not a complete descriptor of image clarity.^{5,6}

**FIGURE 17-2**

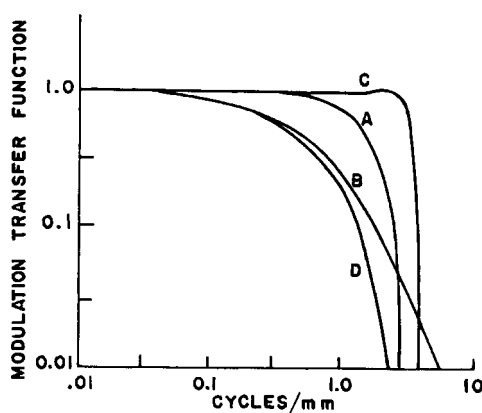
MTF curves for components of a radiographic imaging system and the composite MTF for the entire system. **A:** MTF for the screen–film combination. **B:** MTF for a 1-mm focal spot with 90 cm between the focal spot and the object and 10 cm between the object and film. **C:** MTF for 0.3-mm motion of the object during exposure. **D:** Composite MTF for the entire imaging system. (From Morgan, R.⁷)

The MTF of any imaging system is a product of the MTFs of the individual components of the system. That is, the MTF of the complete system can be computed if the MTFs of the individual components are known. This principle is illustrated in Example 17-1.

Example 17-1

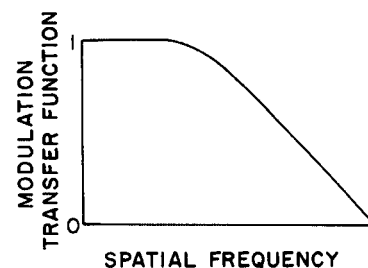
At a spatial frequency of 5 cycles per millimeter, the MTFs of components of an x-ray film–screen imaging system are focal spot, 0.9; motion, 0.8; and intensifying screen, 0.7. The MTF of the composite imaging system is $0.9 \times 0.8 \times 0.7 = 0.5$.

Example 17-1 illustrates the rule that the MTF of a composite imaging system at any particular spatial frequency is always less than the MTF of the system component with the poorest resolution at that frequency. This rule is illustrated in Figure 17-2, where the composite MTF curve for a radiographic imaging unit is below the MTF curves for the components of the imaging system at every spatial frequency. This is shown also in the MTF curves for the fluoroscopic system shown in Figure 17-3.

**FIGURE 17-3**

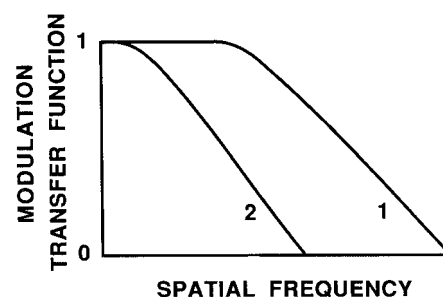
MTF curves for components of a fluoroscopic imaging system and the composite MTF for the system. **A:** MTF for a 1-mm focal spot with a ratio of 3 between the target–object distance and the object–screen distance. **B:** MTF for the image intensifier. **C:** MTF for a 525-line television system. **D:** composite MTF for the entire system. (From Morgan, R.⁸)

A test object of alternating slits and spaces is referred to as a “line pair phantom.”

**MARGIN FIGURE 17-7**

Modulation transfer function (MTF) for a typical imaging system.

The frequency where the MTF is 0.1 is usually taken as the cutoff frequency; frequencies above this value are considered to be unrepresented in the image. The cutoff frequency sometimes is described as the resolving power of the imaging system.

**MARGIN FIGURE 17-8**

MTF curves for two imaging system. Superior spatial resolution is provided by system 1.

The MTF is said to be the representation of the PSF or LSF in “frequency space.”

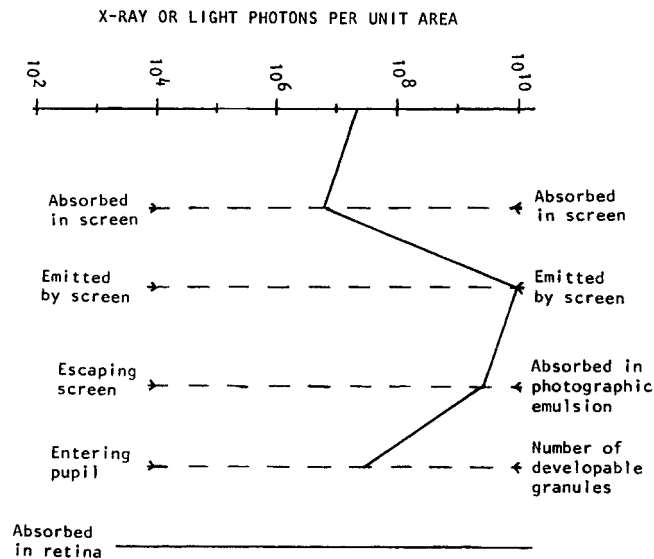


FIGURE 17-4
Quantum levels for stages of information conversion in film-screen radiography.

■ QUANTUM LEVELS AND CONVERSION EFFICIENCIES

The number of x rays or light photons per unit area may be described at each stage of transformation of information in an x-ray beam into a visual image. These numbers are termed *quantum levels*. For example, a typical radiograph may require an exposure of 3×10^7 x-ray photons per square centimeter to the intensifying screen (Figure 17-4). About 30% of the photons may be absorbed in the screen. For each x ray absorbed, 1000 photons of light may be liberated. About 50% of these light photons escape from the screen and interact in the emulsion of the x-ray film. In a typical photographic emulsion, 200 photons of light are absorbed for each photographic granule affected. In the data in Figure 17-4 the number of photons per unit area is lowest at the stage of absorption of x-ray photons in the intensifying screen. This stage is termed the *quantum sink*. The quantum sink identifies the stage of greatest influence of quantum mottle on image resolution.

■ SUMMARY

- The point spread function is a mathematical expression that describes the ability of an imaging system to reproduce a point object (point response).
- The line spread function is a mathematical expression that describes the ability of an imaging system to reproduce a line object (line response).
- The edge spread function is a mathematical expression that describes the ability of an imaging system to reproduce an edge (edge response).
- The contrast response function is a mathematical expression that describes the amplitude of response in the image as a function of spatial frequency (contrast response).
- The modulation transfer function is a mathematical expression that describes the ability of an imaging system to capture and display modulations present in the radiation beam from the object.
- The stage in the imaging process where the number of information carriers is lowest is the quantum sink.

PROBLEMS

- *17-1. The space between two slits in an x-ray-opaque lead sheet can be narrowed to 0.2 mm before the slits are no longer distinguishable in an image. Determine the resolution distance and the cutoff frequency of the imaging system.
- *17-2. In a radiographic unit, would the cutoff frequency be higher or lower on the anode side versus the cathode side of the x-ray beam? Why?
- *17-3. A fluoroscopic imaging system has the following component MTFs at 3 line pairs per millimeter: focal spot, 0.9; motion, 0.7; input screen, 0.9; intensifier electronics, 1.0; output screen, 0.9; optics, 0.9; and television system, 0.9. What is the composite MTF of the system?

*For those problems marked with an asterisk, answers are provided on p. 492.

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VISUAL PERCEPTION

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Paintings in caves of southern France and northern Spain are thought to reflect attempts by early humans to “visualize” events such as a successful hunt before embarking on the experience.

Visual perception has been the subject of philosophical speculation for centuries, beginning with the Greeks and probably before. Aristotle suggested that a unified perception of an object occurred through the exercise of the faculty of “common sense” that resides in the heart.¹

Euclid and Ptolemy developed a geometric theory that suggested that the eye perceives an object by projecting something toward that object. Aristotle and the atomists believed the reverse. The argument was settled more than 800 years later when in 965 an Arabian scientist in Basra (now Iraq), Abu Ali al-Hasan Ibn al-Haytham, convinced some observers to stare at the sun until their eyes became damaged. This was accepted as evidence that something must enter the eye from the outside. While of questionable ethics, this event marked a transition in philosophy from complete faith in theory and conjecture to an evolving acceptance of the primacy of experimental evidence.

Three stages of visual perception:

- Detection
- Recognition
- Interpretation

The frequency of missed signals during interpretation of radiographs has not changed much in spite of major advances in equipment and technique.

Variations in Detection of Radiologic Information

Miss rate	20–30%
Interobserver variation	10–20%
Intraobserver variation	5–10%

OBJECTIVES

After completing this chapter, the reader should be able to:

- Distinguish between detection, recognition and interpretation of information in an image.
- Describe the anatomy and physiology of the human eye.
- Explain various models for detecting visual signals in images.
- Delineate the concepts of visual acuity and contrast discrimination.
- Identify and explain various expressions for describing visual performance, including
 - Contrast-detail analysis
 - Conspicuity
 - Truth tables
 - Relative operating characteristics

INTRODUCTION

Visual images have always been essential to the efforts of humans to understand, characterize, and accommodate to their environment. Long before written communication, images were used to convey information and to express inner thoughts and desires.

Today, visual images serve in many ways to display and communicate complex information in a highly sophisticated world society. Economic transactions, transportation control, security and surveillance, and process control in industry depend on images, as do almost all interactions among individuals and groups. In the area of human services, medical diagnosis and treatment are among the specialties most dependent on images. Radiology is the medical discipline primarily responsible for producing and interpreting medical images. Consequently, an understanding of the detection and interpretation of visual information is an essential part of the radiologist’s education.

The perception of visual information consists of the three sequential processes illustrated in Figure 18-1. The first process is the *detection* of visual signals by the observer (Figure 18-2). If visual information is not detected, it cannot be integrated into the perceptual process for its correct interpretation. Important information sometimes is overlooked in radiologic images for reasons that are not well understood. Experiments^{4–7} have revealed that radiologists miss 20% to 30% of the visual clues present in diagnostic images. In 10% to 20% of instances, radiologists detect different signals (interobserver variation) in the same images. Even disagreement with a previous reading by the same radiologist (intraobserver variation) occurs in 5% to 10% of images. These undetected signals and inconsistencies reflect the tendency of all observers to miss important visual information present in images.

Even if visual signals are detected, they may be ignored because they are considered inconsequential. The tendency to dismiss information connotes a problem of *recognition*. The observer simply does not appreciate the importance of the information. Eye tracking experiments performed on radiologists reading diagnostic images have revealed that radiologists frequently look at visual clues that are important to the diagnosis but fail to incorporate them into the interpretive process. This failure indicates that the visual clues are detected but not recognized.

Visual signals may be detected and their importance recognized, but still the diagnosis reached by the radiologist may be incorrect. This error occurs in the *interpretation* phase of information gathering and processing (Figure 18-3), the third step in the model of radiologic diagnosis. Error at this level may reflect inadequate training, inexperience, prior conditioning, fatigue, poor judgment, or simply a difficult case in which the wrong option is chosen.

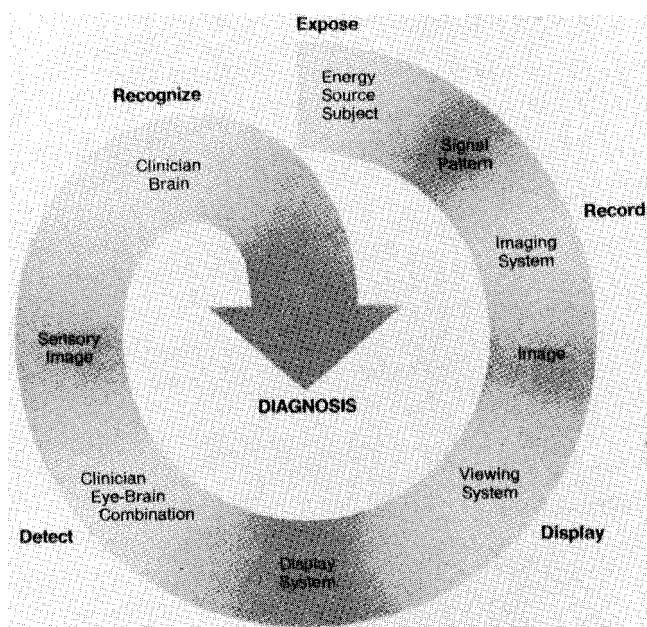


FIGURE 18-1
The perception of visual information. (From Jaffe, C. C.² Used with permission.)

Reasonably good models exist to describe the process of detecting visual signals. Paradigms to explain how these signals are recognized as normal and abnormal, important or unimportant, and expected or unexpected are more rudimentary. The greatest enigma of all is understanding how detected and recognized visual signals are woven into an interpretation that leads to a correct medical diagnosis of the patient's condition.⁹

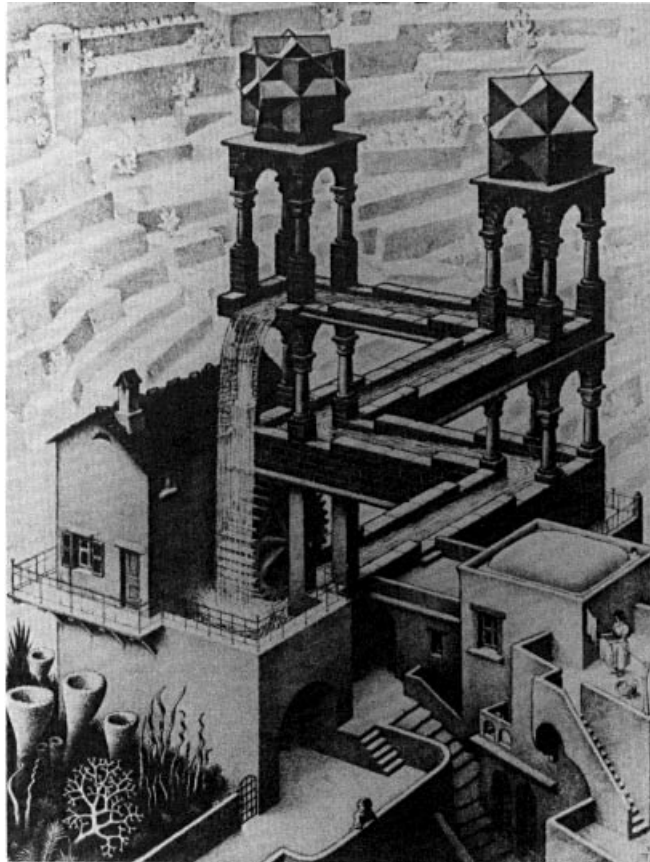


FIGURE 18-2
Example of a detection challenge; Dalmatian dog. (From Thurston, J. B., and Carraher, R. G.³ Used with permission.)



MARGIN FIGURE 18-1
Example of recognition challenge; the observer should be able to recognize both a young and an old lady, although not simultaneously.

For centuries, philosophers assumed that visual artists (e.g., painters and sculptors) experienced a richly pictorial inner life of the mind that they were able to project into their art. In the last century, efforts of psychologists and psychiatrists to probe and characterize this inner life were largely discredited by logical positivists, behavior scientists, and some postmodernist philosophers who believed that people are biological machines that react to external stimuli rather than to thoughts, feelings, and intentions. Recent developments in philosophy, psychology, linguistics, and computer science are restoring interest in the inner life of the mind.

**FIGURE 18-3**

Example of interpretation challenge; moving water without origin. (From Escher, M. C.⁸ Used with permission.)

■ HUMAN VISION

Images are the product of the interaction of the human visual system with its environment. The process of human vision is outlined here; more detailed treatments are available in texts on human physiology.

Anatomy and Physiology of the Eye

The human eye is diagramed in Figure 18-4. It is an approximate sphere that contains four principal features: the cornea, iris, lens, and retina. The retina contains photoreceptors that translate light energy into electrical signals that serve as nerve impulses to the brain. The other three components serve as focusing and filtering mechanisms to transmit a sharp, well-defined light image to the retina.

The wall of the eye consists of three layers (tunics) that are discontinuous in the posterior portion where the optic nerve enters the eye. The outermost tunic is a fibrous layer of dense connective tissue that includes the cornea and the sclera. The cornea comprises the front curved surface of the eye, contains an array of collagen fibers and no blood vessels, and is transparent to visible light. The cornea serves as a coarse focusing element to project light onto the observer's retina. The sclera, or white of the eye, is an opaque and resilient sheath to which the eye muscles are attached. The second layer of the wall is a vascular tunic termed the uvea. It contains the choroid, ciliary body, and iris. The choroid contains a dense array of capillaries that supply blood to the tunics. Pigments in the choroid reduce internal light reflection that otherwise would blur the images. The ciliary body contains the muscles that support and focus the lens. It also

The first known description of eye physiology was provided by Felix Platter of Basle. Others who contributed knowledge about how the human eye functions include Johannes Kepler, Rene Descartes, and Isaac Newton.

Although only about 24 mm in diameter, the eye contains several hundred million working parts that provide a vast range of responses to contrast, color, detail, form, and space. The eye is one of the most complex organs in the body.

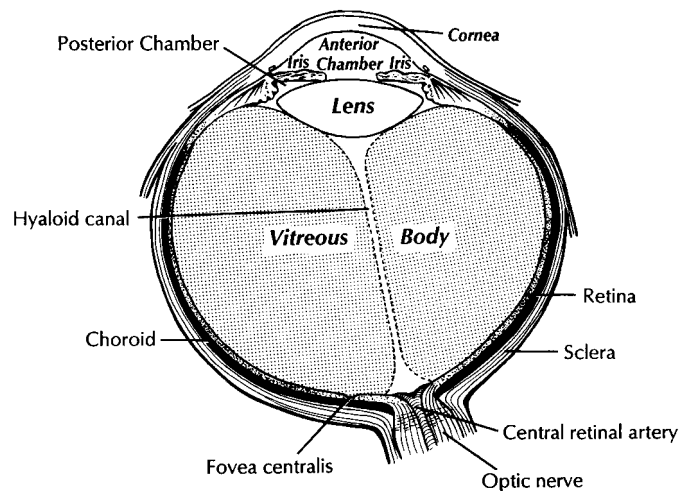


FIGURE 18-4
Horizontal section through the human eye.

contains capillaries that secrete fluid into the anterior segment of the eyeball. The iris is the colored part of the eye that has a central aperture termed the pupil. The diameter of the aperture can be altered through the action of muscles in the iris to control the amount of light entering the eye. The aperture can vary from about 1.5 mm to 8 mm.

The anterior and posterior chambers of the eye are filled with fluid. The anterior chamber contains aqueous humor, a clear plasma-like fluid that is continually drained and replaced. The posterior chamber is filled with vitreous humor, a clear viscous fluid that is not replenished. The cornea, the aqueous and vitreous humors, and the lens serve collectively as the refractive media of the eye.

The lens of the eye provides the fine focusing of incident light onto the retina. It is a convex lens that can change in thickness by action of the ciliary muscles. The index of refraction of the lens is close to that of the surrounding fluids in which it is suspended, so it serves as a fine-focusing adjustment to the coarse focusing function of the cornea. The process of accommodation by which near objects are brought into focus is achieved by contraction of the ciliary muscles. This contraction causes the elastic lens to bow forward into the aqueous humor, thereby increasing its thickness. Accommodation is accompanied by constriction of the pupil, which increases the depth of field of the eye.

With age the lens loses its flexibility and is unable to accommodate so that near objects can be focused onto the retina. This is the condition of presbyopia in which reading glasses are needed to supplement the focusing ability of the lens. Clouding of the lens with age causes a diminution in the amount of light reaching the retina. This condition is known as a lens cataract, which, when severe enough, makes the individual a candidate for surgical replacement of the lens, often with an artificial lens.

The innermost layer of the eye is the retina, which is composed of two components, an outer monolayer of pigmented cells and an inner neural layer of photoreceptors. Because considerable processing of visual information occurs in the retina, it often is thought of more as an outpost of the brain rather than as simply another component of the eye. There are no photoreceptors where the optic nerve enters the eye, creating a blind spot. Near the blind spot is the macula lutea, an area of about 3 mm² over which the retina is especially thin. Within the macula lutea is the fovea centralis, a slight depression about 0.4 mm in diameter. The fovea is on the optical axis of the eye and is the area where the visual cones are concentrated to yield the greatest visual acuity.

The retina contains two types of photoreceptors, termed rods and cones. Rods are distributed over the entire retina, except in the blind spot and the fovea centralis. The retina contains about 125 million rods, or about 10⁵/mm². Active elements in the rods

The cornea contributes about two-thirds of the refracting power of the eye (refracting power is the ability to change the direction of incident light). The remaining one-third is contributed principally by the surfaces of the lens.

Underwater vision without a mask is blurred and hazy because the eye is bathed by water of about the same refractive index as the cornea. Hence, the refracting power of the cornea is greatly diminished, and the retinal image is unfocused.

Accommodation of the lens is similar in function to the auto-focus capacity of a camera.

If an image is not projected in focus on the retina, the eye is said to be *ametropic*. There are three types of ametropia: myopia (short-sightedness), hyperopia (long-sightedness), and astigmatism (lack of symmetry on the cornea and lens surfaces).

Cataracts and other lens defects produce the vision problem known as aphakia.

The retina does not “record” an image projected onto it by the lens. Instead, it responds to the incident image by sending coded neural signals to the brain, where they are decoded to yield the sense of visual perception. Hence, the optics of the eye do not have to be as perfect as those of an optical instrument. Hermann von Helmholtz, the nineteenth-century physicist who developed the field of physiological optics, said that if an optician built so imperfect an optical instrument as the human eye, he would undoubtedly become bankrupt.¹⁰

It is often said that rods encode the twilight picture, and cones encode the daylight picture.

The process of viewing with rods is known as “scotopic” vision. Viewing with cones is known as “photopic” vision.

The physiologist Lord Adrian said that the “copy” furnished by the >100 million reporters (rods and cones) is severely edited and condensed, and only the more dramatic features are telegraphed in code from the retina to the central offices in the brain.¹⁰

One popular theory (the trichromatic theory) of color vision proposes that three types of cones exist, each with a different photosensitive pigment that responds maximally to a different wavelength (450 nm for “blue” cones, 525 nm for “green” cones, and 555 nm for “red” cones). The three cone pigments share the same chromophore as the rods; their different spectral sensitivities result from differences in the opsin component.

In a now famous prism experiment, Newton showed in 1666 that sunlight consists of a mixture of visible light rays, each with its own color. Goethe responded to this finding by stating that it was absurd to assert that mere mixing of rainbow colors could appear white, since white is without color.

The spectral response of human vision peaks near the center of the sun’s spectrum under conditions of bright illumination. The peak response shifts toward the blue end of the visible spectrum at twilight, in order to match the spectrum of scattered light in the sky.

The detection threshold of the human eye is often said to be the light from a single candle on a clear, dark night at a distance of 30 miles.

An alternative expression for the relationship between stimulus intensity I and the visual perception S of the stimulus is known as Stevens’ power law: $S = kI^n$, where k is a constant and n is the power.

In 1860 Gustav Fechner introduced the term *psychophysics* to characterize the study of the perceptions of stimuli and their relationship to the physical characteristics of the stimuli.

(and in the cones as well) are replenished throughout an individual’s lifetime. Rods have a low but variable threshold to light, and they respond to very low intensities of incident light. Vision under low illumination levels (e.g., night vision) is attributable almost entirely to the rods.

Rods contain the light-sensitive pigment rhodopsin (visual purple) which undergoes a chemical reaction (the rhodopsin cycle) when exposed to visible light. Rhodopsin consists of a lipoprotein called opsin and a chromophore (a light-absorbing chemical compound called 11-*cis*-retinal).¹¹ The chemical reaction begins with the breakdown of rhodopsin and ends with the recombination of the breakdown products back into rhodopsin. The recovery process takes 20–30 minutes, which is the time required to accommodate to low levels of illumination (dark adaptation). The rods are maximally sensitive to light of about 510 nm, in the blue-green region of the visible spectrum. Rods cannot discriminate different wavelengths of light, and vision under low illumination conditions is essentially “colorblind.” More than 100 rods are connected to each ganglion cell, and the brain cannot discriminate among these photoreceptors to identify the origin of an action potential transmitted along the ganglion. Hence, rod vision yields relatively low visual acuity, in combination with a high sensitivity to low levels of ambient light.

The retina contains about 7 million cones, packed tightly in the fovea and diminishing rapidly across the macula lutea. The density of cones in the fovea is about 140,000/mm². Cones are maximally sensitive to light of about 550 nm, in the yellow-green region of the visible spectrum. Cones are much ($1/10^4$) less sensitive than rods to light, but in the fovea there is a 1:1 correspondence between cones and ganglions, so the visual acuity is very high. Cones are responsible for color vision through mechanisms that are imperfectly understood at present.

Properties of Vision

For two objects to be distinguished on the retina, light rays from the objects must define at least a minimum angle as they pass through the optical center of the eye. The minimum angle is defined as the visual angle. The visual angle, expressed in units of minutes of arc, defines the visual acuity of the eye. An individual with excellent vision and who is functioning in ideal viewing conditions can achieve a visual angle of about 0.5 minutes of arc, which is close to the theoretical minimum defined by the packing density of cones on the retina.¹²

The eye is extremely sensitive to small amounts of light. Although the cones do not respond to illumination levels below a threshold of about 0.001 cd/m², the rods are much more sensitive and respond to just a few photons. For example, as few as 10 photons can generate a visual stimulus in an area of the retina where the rods are present at high concentration.¹³

Differences in signal intensity that can just be detected by the human observer are known as just noticeable differences (JND). This concept applies to any type of signal, including light, that can be sensed by the observer. The smallest difference in signal that can be detected depends on the magnitude of the signal. For example, the brightness difference between one and two candles may be discernible, whereas the difference between 100 and 101 candles probably cannot be distinguished. This observation was quantified by the work of Weber, who demonstrated that the JND is directly proportional to the intensity of the signal. This finding was quantified by Fechner as

$$dS = k[dI/I]$$

where I is the intensity of stimulus, dS is an increment of perception (termed a *limen*), and k is a scaling factor. The integral form of this expression is known as the Weber–Fechner law:

$$S = k[\log I/I_0]$$

The Weber–Fechner law is similar to the expression for the intensity of sound in decibels, and it provides a connection between the objective measurement of sound intensity and the subjective impression of loudness.¹⁴

DETECTION OF VISUAL INFORMATION

Various models have been developed to describe how observers detect visual signals in images. These models can be categorized into three approaches to understanding visual detection: signal-to-noise, computational, and signal channel models.

Signal-to-Noise Model

The signal-to-noise model describes the ability of an observer to detect simple visual signals embedded in a noisy background. Noise in the image (e.g., quantum mottle, extraneous structures, and receptor artifacts) is referred to as external noise.¹⁵ The model characterizes an “ideal observer” who detects signals with a certain likelihood whenever their amplitude differs from background by some prescribed threshold, usually a multiplicative factor of 2 to 5. The inability of a real observer to detect visual signals as well as the ideal observer is said to be caused by “internal noise” in the real observer. Sources of internal noise include distraction of the observer, preconditioning of the observer by the presence or absence of signals in previously viewed images, reduced intrinsic sensitivity to visual signals in some observers, and a myriad of other possible causes.

The performances of ideal and actual observers are compared in the margin for the simple case in which actual observers are asked to state whether a simple signal is present or absent in an image. The index of detectability on the ordinate (y axis) is the inverse of the size of subtle lesions that can be detected with reasonable assurance. As shown in the illustration, the performance of the human observer usually falls short of that for the ideal observer. This difference may be due in part to selection by the human observer of a visual threshold different from that assumed for the ideal observer. It may also be caused in part by internal noise (unpredictability) in the observer.

Wagner and Brown¹⁶ have suggested that the performance of an imaging system can be characterized quantitatively by describing the ability of an ideal observer to detect simple visual signals provided by the system. They have proposed this approach for evaluation and comparison of imaging systems used in clinical radiology.

Computational Model

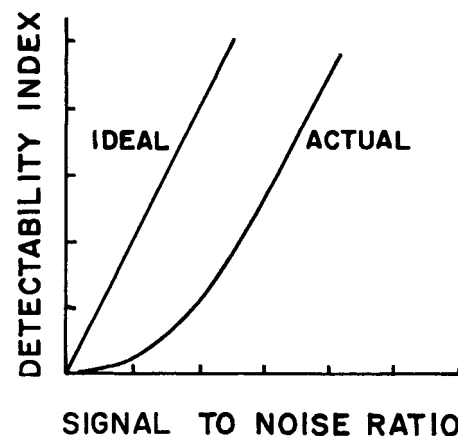
The computational model of visual detection has evolved from experience in designing computer programs to extract information from images. The model proposes that the first step in detecting visual information involves the rapid identification of brightness (density) changes in the image, together with properties such as orientation, location, depth, and spatial frequency of these changes. This “primal sketch” of the image is obtained by a simple psychophysical reaction to the image without use of *a priori* knowledge of the structure and function of the object represented in the image. That is, visual signals are extracted from the image without any assumptions about the nature of the image and the type of visual scene portrayed.¹⁷ Initial applications of the computational model were promising; however, its development into a useful paradigm for visual detection has been disappointing. Probably our current understanding of psychophysical properties of human vision is simply too rudimentary to develop a satisfactory computational model for the visual detection process.¹⁸

Signal-Channel Model

In this model of visual detection, the human visual system is characterized as a series of parallel vision channels, with a specific frequency range of visual information assigned to each channel. Several studies suggest that the frequency range of each channel is on the order of ± 1 octave of spatial frequency information¹⁹ and that the visual system as a whole functions essentially as a frequency analyzer.

Development of the signal channel model has been aided by the use of sinewave gratings, a series of light and dark bars repeated at increasing frequency (Figure 18-5).

Both the Weber–Fechner law and the Stevens power law have been shown to be seriously deficient under certain experimental conditions. These findings reveal that the ability of humans to quantify sensations such as brightness and loudness is flawed. Consequently, we remain dependent on measuring devices such as light meters and speedometers.



MARGIN FIGURE 18-2

The visual performance of actual and “ideal” observers in a lesion detectability experiment.

An experiment where the participant is required to select one of two choices is called a two-alternative, forced-choice test.

Studies of the definition and communication of information, and its separation from background “noise,” constitute the discipline of Information Theory. This discipline was founded in the late 1940s by the engineer Claude Shannon and the mathematician Norbert Wiener.

The computational model suggests that visual detection consists of two stages: (1) a preattentive (early warning) stage for almost instantaneous detection of textural changes in the environment and (2) an attentive stage for focusing attention on specific objects identified in the preattentive stage.¹

Although the computational model is limited in its ability to fully characterize human vision, it has proven to be exceptionally useful in the design of reconnaissance systems and pattern recognition software programs in artificial intelligence.

A person who has excellent visual acuity (e.g., 20/20 with the Snellen test), but has impaired contrast vision, may be unable to see a truck in the fog or read a low-contrast x-ray image.



MARGIN FIGURE 18-3
Snellen eye test chart.

If the lettering on a Snellen chart is reversed (i.e., white letters on a black background), the ability of observers to recognize the letters from a distance is greatly impaired.

Magnification and minification lenses, along with viewing images from different distances, are useful techniques in radiology because they help match image detail to the spatial-frequency sensitivity of human vision (1–5 cycles per degree).

The ability of the human eye to filter spatial frequencies explains the usefulness of certain image-processing techniques such as edge enhancement, gray-level quantization, and color mapping.

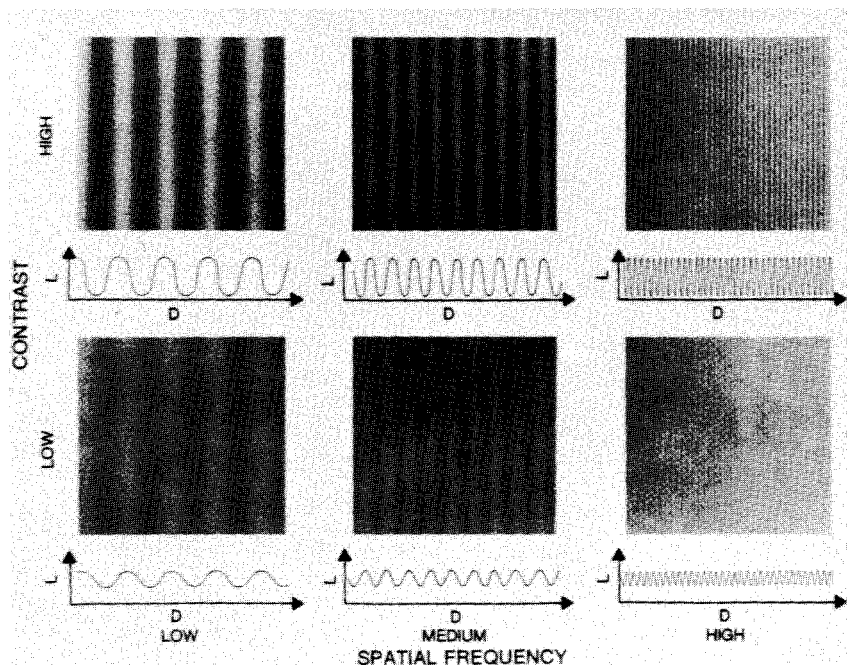


FIGURE 18-5

Sine-wave gratings used to determine the sensitivity to visual information of different spatial frequencies. (From Ginsburg, A. P.²¹ Used with permission.)

Images of these gratings can be used as test objects to evaluate the ability of the observer to extract visual signals from images of increasing complexity.^{20,21}

VISUAL ACUITY

Visual acuity describes the ability of an observer to extract detailed information from a small region of an image. In tests of visual acuity, high-contrast images are used to ensure that results are not compromised by the inability of the observer to distinguish subtle differences in brightness (density) in the image. Although visual acuity can be evaluated in several ways, the most common method is use of the Snellen eye chart shown in the margin. The Snellen chart consists of rows of letters that diminish in size from top to bottom. Results of Snellen tests are often expressed as the ratio of the maximum distance that an observer can just see an object divided by the distance that most people can see the same object. A measure of visual acuity of 20/40 simply means that the individual barely sees letters at 20 ft that the average person would see at 40 ft.

Persons are frequently tested for visual acuity, and corrective eyewear is designed principally to compensate for deficiencies in this property. Yet visual acuity is only part of the answer to whether people can detect visual information. Another important feature is the ability of the observer to distinguish subtle changes in brightness (density) in the image.

CONTRAST DISCRIMINATION

Contrast discrimination is the ability of the visual system to distinguish differences in brightness (density) in the image. This ability is described by a contrast sensitivity curve (Figure 18-6). This curve reveals the tendency of an observer to filter spatial frequencies so that coarse and fine details of the image are suppressed as compared with information of intermediate spatial frequencies.²²

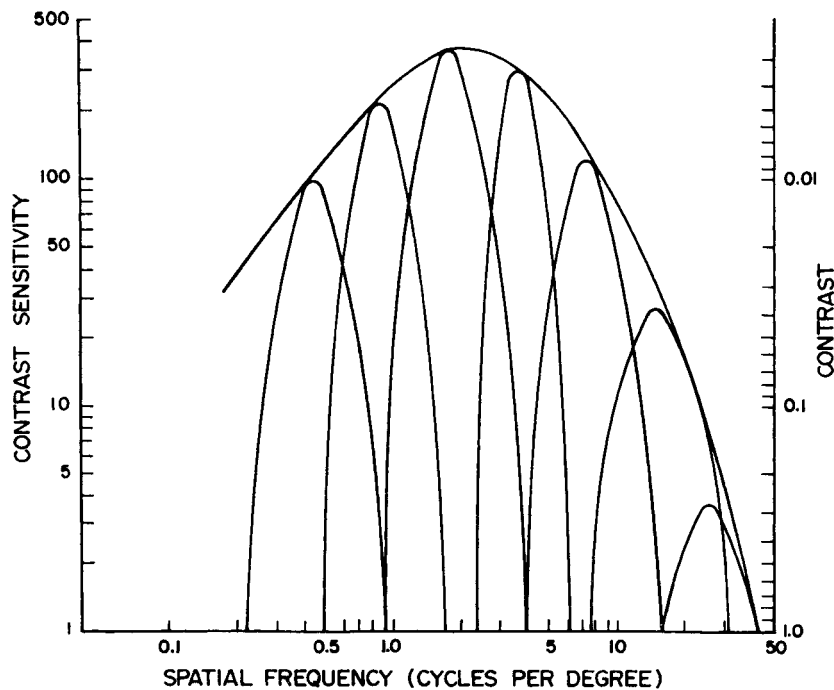


FIGURE 18-6

A contrast sensitivity curve. (From Ginsburg, A. P.²¹ Used with permission.)

Contrast discrimination can be evaluated in different ways. One approach uses the vision contrast test system²³ depicted in Figure 18-7. Contrast discrimination is probably a more critical feature than visual acuity in determining how well the average person “sees.” Determination of contrast discrimination has proved helpful in evaluating the extent of eye diseases such as cataracts, glaucoma, amblyopia, and the visual consequences of diabetes.

Vision loss at high frequencies, as detected with the Snellen test, is usually indicative of an optical (blurring) problem. Vision loss at middle and low frequencies, with vision at the higher frequencies intact, suggests a neuro-ophthalmological cause.

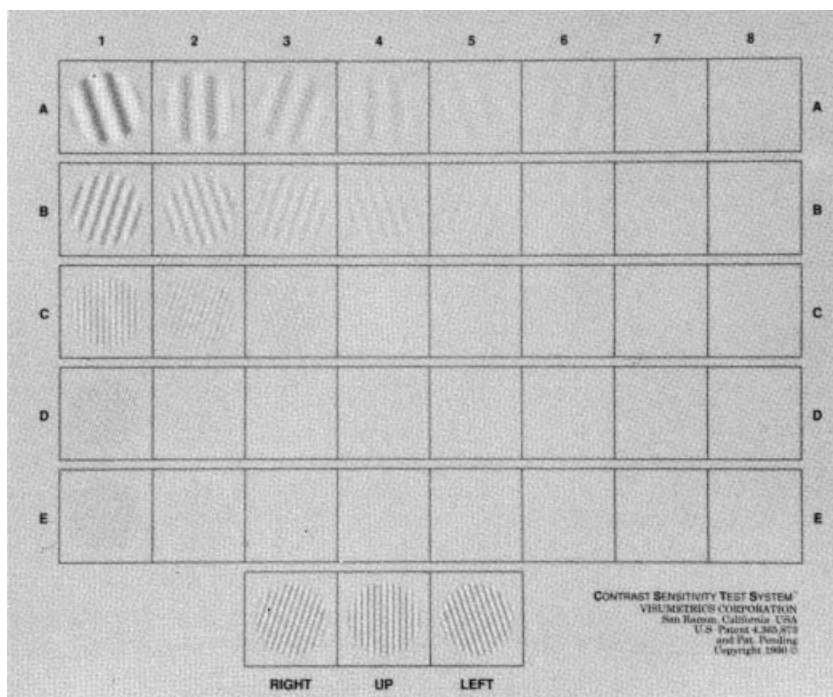


FIGURE 18-7

Vision contrast test system. (Courtesy of A. P. Ginsburg.)

Tests of contrast sensitivity have proven to predict the ability to detect and recognize letters, faces, aircraft in radar simulators, and road signs.²¹

Some authors have suggested that contrast discrimination should be evaluated whenever contact lenses and spectacles are fitted. A few²⁴ have even suggested that contrast discrimination might be a useful test in selecting candidates for occupations strongly dependent on image interpretation. Among these occupations might be aircraft pilots, radar operators, surveillance experts, and radiologists. Candidates could be selected by criteria that include matching the contrast discrimination characteristics of the individual to the contrast range and distribution presented in the images to be interpreted. In radiology, this suggestion has not been enthusiastically received.

■ RECOGNITION AND INTERPRETATION OF VISUAL INFORMATION

Radiologic diagnosis is a more complex process than simply detecting visual signals in a relatively uniform background. How these signals are recognized and interpreted and how this interpretation is woven into a radiologic diagnosis are poorly understood phenomena. Several vision scientists have proposed that an experienced observer first looks quickly at an image and compares it with a template of “remembered” images that represent normal structures similar to those in the image. In effect, the observer performs a quick mental subtraction to leave only those items in the perceived image that differ from the normal template. These differences are then examined more closely, and decisions are made about their significance. If the observer decides that they are acceptable deviations from normal, they receive no further attention. But if they are judged to be unusual, they are integrated into an interpretation of the image that reflects an abnormal condition in the patient. This “cognitive” model of image perception is a tempting explanation of the way visual information is detected, recognized, and interpreted. However, several experiments with human observers suggest that it is overly simplistic.^{25–28}

Studies of visual search patterns reveal that observers inexperienced in the types of images presented (e.g., chest radiographs) tend to focus on the central part of the image, whereas more experienced observers employ a more circumferential and complete scanning pattern.

■ EXPRESSIONS OF VISUAL PERFORMANCE

Visual images are useful insofar as they convey aesthetically pleasing or helpful information to the observer. In the latter case, helpfulness is characterized by greater enlightenment and improved performance of the observer. Several approaches have been developed to express these characteristics in a quantitative manner. Some of the more common approaches are described in this section.

Contrast-Detail Analysis

The detection of simple objects in an image is a complex interplay of their size, brightness (density), and differences from background. This interplay can be represented in a contrast-detail diagram. In this approach, a test pattern is constructed that contains objects of different sizes and composition superimposed on a relatively uniform background. The objects of different composition are reflected as regions of different brightness (density) in the image. The observer is asked to identify the smallest object of each brightness or density that is visible in the image. Smaller objects must be displayed at greater brightness or density to be seen in the image.

Conspicuity

The detection of objects in an image is complicated by the presence of background information, especially when the background has an intrinsic structure of its own. In studies of this complicating factor, a description of the relationship of the objects and background is helpful. One useful description is conspicuity, defined as the contrast between the objects and background divided by the average complexity of